

A 5^r -Scaled Family of the Reduced Ternary $(5k \pm 1)/4$ Collatz-Type Map

Exact Algebraic Conjugacy, Cycle Preservation, Orbit-Statistic Invariance, and Transfer of the 10^{10} Finite Verification

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Abstract

We develop the full 5^r -scaled family associated with the residue-dependent reduced ternary $(5k \pm 1)/4$ Collatz-type map studied in the author's preceding work. The base state space is the positive odd integers and the base affine numerator is

$$M_0(y) = \begin{cases} 5y - 1, & y \equiv 1 \pmod{4}, \\ 5y + 1, & y \equiv 3 \pmod{4}, \end{cases} \quad T_0(y) = \frac{M_0(y)}{2^{\nu_2(M_0(y))}}.$$

For every integer $r \geq 0$ we define the scaled domain $S_r = 5^r \mathbb{N}_{\text{odd}}$ and

$$M_r(x) = \begin{cases} 5x - 5^r, & x \equiv 1 \pmod{4}, \\ 5x + 5^r, & x \equiv 3 \pmod{4}, \end{cases} \quad T_r(x) = \frac{M_r(x)}{2^{\nu_2(M_r(x))}}.$$

Thus the unreduced branches have the requested form $(5k \pm 5^r)/4$ after the forced factor 4 is removed, followed, as in the base system, by complete removal of any additional powers of 2.

The central result is an exact algebraic conjugacy. Since $5^r \equiv 1 \pmod{4}$ and $\gcd(5^r, 2) = 1$, multiplication by 5^r preserves both the branch residue and the complete two-adic valuation. Consequently

$$T_r(5^r y) = 5^r T_0(y), \quad T_r^j(5^r y) = 5^r T_0^j(y) \quad (j \geq 0).$$

Hence every orbit relation, branch sequence, valuation sequence, periodic orbit, minimal cycle length, capture time, first-descent stopping time, basin membership, and normalized orbit geometry is transported bijectively between the base system and its scaled copy. Absolute reduced and raw affine peaks are multiplied by 5^r , while peak/start ratios are unchanged. In particular, the base fixed point 1 becomes 5^r , and the base three-cycle $7 \rightarrow 9 \rightarrow 11 \rightarrow 7$ becomes

$$7 \cdot 5^r \rightarrow 9 \cdot 5^r \rightarrow 11 \cdot 5^r \rightarrow 7 \cdot 5^r.$$

No additional periodic orbit can occur inside S_r unless a corresponding base periodic orbit exists.

The preceding base paper exhaustively verified all 5,000,000,000 positive odd starts below 10^{10} . We do not relabel that computation as a new scan. Instead, the conjugacy theorem transfers each verified base trajectory exactly to the corresponding scaled set

$$\{5^r y : 1 \leq y < 10^{10}, y \text{ odd}\} \subset S_r,$$

which reaches absolute scale below $5^r 10^{10}$. The global two-attractor convergence statement remains open: the present theorem proves exact equivalence of the scaled and base dynamical problems, not global convergence of the unresolved base problem.

Keywords: Collatz-type map; $5x \pm 1$; $(5k \pm 5^r)/4$; 5^r scaling; algebraic conjugacy; two-adic valuation; periodic orbits; cycle preservation; computational number theory.

1 Introduction

The reduced $5x \pm 1$ map considered in the author's base paper is a residue-dependent odd-to-odd dynamical system. For an odd state y , the sign is chosen from $y \bmod 4$ so that $5y \pm 1$ is divisible by at least 4, after which all powers of 2 are removed. The base study developed closure, an exact periodic-orbit identity, a two-adic valuation law and drift heuristic, inverse branches and basin structure, and an exhaustive finite verification through 10^{10} .

The present paper asks a structural rather than a new billion-scale computational question. Replace the constants ± 1 by $\pm 5^r$ and simultaneously restrict the state space to the 5^r -scaled copy of the base odd domain. Does this create genuinely new dynamics? We prove that it does not. For every integer $r \geq 0$, the resulting system is exactly conjugate to the base map through the linear scaling

$$\Phi_r(y) = 5^r y.$$

This is stronger than numerical similarity for selected r . Exact conjugacy proves the correspondence for every $r \geq 0$ at once and identifies precisely what is preserved: full trajectories, branch choices, complete two-adic valuations, cycle lengths, transient/capture times, first-descent stopping times, basin membership, and all normalized orbit ratios. Only dimensional quantities such as state values and peaks are multiplied by 5^r .

The finite computation in the base article remains base-system evidence. Its numerical counts are inherited here through a theorem and are not presented as fresh scans of infinitely many scaled systems. This distinction is essential for reproducibility and for separating exact algebraic consequences from finite computational evidence.

2 The base reduced ternary map

Let

$$S = \mathbb{N}_{\text{odd}} = \{1, 3, 5, \dots\}.$$

Define the sign selector

$$\varepsilon(y) = \begin{cases} -1, & y \equiv 1 \pmod{4}, \\ +1, & y \equiv 3 \pmod{4}. \end{cases}$$

The base affine numerator and reduced map are

$$M_0(y) = 5y + \varepsilon(y), \quad T_0(y) = \frac{M_0(y)}{2^{\nu_2(M_0(y))}}. \quad (1)$$

Because $5 \equiv 1 \pmod{4}$, one has $4 \mid M_0(y)$ for every odd y , so $\nu_2(M_0(y)) \geq 2$ and $T_0(y)$ is again odd.

Two exact periodic attractors displayed in the base work are

$$1 \rightarrow 1$$

and

$$7 \rightarrow 9 \rightarrow 11 \rightarrow 7,$$

since $5 \cdot 7 + 1 = 36 = 4 \cdot 9$, $5 \cdot 9 - 1 = 44 = 4 \cdot 11$, and $5 \cdot 11 + 1 = 56 = 8 \cdot 7$.

3 The 5^r -scaled family

Fix an integer $r \geq 0$.

3.1 Scaled domain

Define

$$S_r = 5^r S = \{5^r y : y \in S\}. \quad (2)$$

Every $x \in S_r$ has a unique representation $x = 5^r y$ with y odd. Since

$$5^r \equiv 1 \pmod{4},$$

we have

$$5^r y \equiv y \pmod{4}. \quad (3)$$

Thus the scaling preserves the branch selector exactly:

$$\varepsilon(5^r y) = \varepsilon(y).$$

3.2 Scaled affine and reduced maps

For $x \in S_r$, define

$$M_r(x) = 5x + 5^r \varepsilon(x) = \begin{cases} 5x - 5^r, & x \equiv 1 \pmod{4}, \\ 5x + 5^r, & x \equiv 3 \pmod{4}. \end{cases} \quad (4)$$

The corresponding completely reduced map is

$$T_r(x) = \frac{M_r(x)}{2^{\nu_2(M_r(x))}}. \quad (5)$$

Because $5^r \equiv 1 \pmod{4}$, the branch rule gives $M_r(x) \equiv 0 \pmod{4}$ for every $x \in S_r$. Therefore the commonly written first compressed form is

$$\frac{5x \pm 5^r}{4},$$

but the dynamical map (5) continues to remove every additional factor of 2, exactly as in the base paper.

Remark 3.1 (Why the scaled domain matters). The exact theorem concerns $S_r = 5^r S$. For a general odd integer x not divisible by 5^r , the factorization used below is unavailable. Therefore “the same dynamics” is asserted on the scaled copy S_r , not on all odd integers under the formula $5x \pm 5^r$.

4 The key two-adic scaling identities

Lemma 4.1 (Affine scaling). *For every $r \geq 0$ and $y \in S$,*

$$M_r(5^r y) = 5^r M_0(y). \quad (6)$$

Proof. By (3), $\varepsilon(5^r y) = \varepsilon(y)$. Hence

$$M_r(5^r y) = 5(5^r y) + 5^r \varepsilon(y) = 5^r (5y + \varepsilon(y)) = 5^r M_0(y).$$

□

Lemma 4.2 (Valuation invariance). *For every nonzero integer n and every $r \geq 0$,*

$$\nu_2(5^r n) = \nu_2(n). \quad (7)$$

Proof. The factor 5^r is odd and therefore contributes no factor of 2. $\square$

Combining the lemmas,

$$\nu_2(M_r(5^r y)) = \nu_2(M_0(y)). \quad (8)$$

Thus corresponding trajectories do not merely scale as sets of states; they remove exactly the same power of 2 at every matched step.

5 Exact algebraic conjugacy

Define the scaling bijection

$$\Phi_r : S \rightarrow S_r, \quad \Phi_r(y) = 5^r y,$$

with inverse $\Phi_r^{-1}(x) = x/5^r$.

Theorem 5.1 (One-step conjugacy). *For every $r \geq 0$ and $y \in S$,*

$$T_r(5^r y) = 5^r T_0(y). \quad (9)$$

Equivalently,

$$T_r \circ \Phi_r = \Phi_r \circ T_0, \quad T_r = \Phi_r \circ T_0 \circ \Phi_r^{-1}.$$

Proof. Using (6) and (8),

$$T_r(5^r y) = \frac{5^r M_0(y)}{2^{\nu_2(M_0(y))}} = 5^r T_0(y).$$

$\square$

Theorem 5.2 (All iterates). *For every $j \geq 0$, $r \geq 0$, and $y \in S$,*

$$T_r^j(5^r y) = 5^r T_0^j(y). \quad (10)$$

Proof. The case $j = 0$ is immediate. If the identity holds for j , then

$$T_r^{j+1}(5^r y) = T_r(5^r T_0^j(y)) = 5^r T_0(T_0^j(y)) = 5^r T_0^{j+1}(y).$$

$\square$

Equation (10) is the organizing identity of the paper. Every statement below follows either immediately from it or from the stepwise valuation identity (8).

6 Exact preservation of periodic orbits

Theorem 6.1 (Cycle bijection). *A tuple $C = (c_0, \dots, c_{L-1})$ is a cycle of T_0 if and only if*

$$5^r C = (5^r c_0, \dots, 5^r c_{L-1})$$

is a cycle of T_r . Minimal cycle length is preserved.

Proof. If $T_0^L(c_0) = c_0$, then

$$T_r^L(5^r c_0) = 5^r T_0^L(c_0) = 5^r c_0.$$

Conversely, if $x = 5^r y \in S_r$ and $T_r^L(x) = x$, division of

$$5^r T_0^L(y) = 5^r y$$

by 5^r gives $T_0^L(y) = y$. Bijectivity of Φ_r preserves minimal period. $\square$

Corollary 6.2. *No periodic orbit exists in S_r that is not the 5^r -multiple of a base periodic orbit. Scaling introduces no new periodic structure inside the scaled domain.*

The displayed base attractors become

$$5^r \rightarrow 5^r \quad (11)$$

and

$$7 \cdot 5^r \rightarrow 9 \cdot 5^r \rightarrow 11 \cdot 5^r \rightarrow 7 \cdot 5^r. \quad (12)$$

For example,

$$\begin{aligned} r = 1 : \quad & 35 \rightarrow 45 \rightarrow 55 \rightarrow 35, \\ r = 2 : \quad & 175 \rightarrow 225 \rightarrow 275 \rightarrow 175, \\ r = 3 : \quad & 875 \rightarrow 1125 \rightarrow 1375 \rightarrow 875. \end{aligned}$$

These are theorem-level consequences, not independent numerical discoveries.

7 The periodic-orbit equation under scaling

For a base orbit write

$$2^{a_i} y_{i+1} = 5y_i + \varepsilon_i, \quad a_i = \nu_2(5y_i + \varepsilon_i) \geq 2.$$

For the scaled orbit $x_i = 5^r y_i$,

$$2^{a_i} x_{i+1} = 5x_i + 5^r \varepsilon_i. \quad (13)$$

Dividing by 5^r recovers the base relation exactly.

Let

$$A = \sum_{i=0}^{L-1} a_i, \quad S_j = \sum_{i=0}^{j-1} a_i, \quad S_0 = 0.$$

Iteration of (13) around a length- L cycle gives

$$(2^A - 5^L)x_0 = 5^r \sum_{j=0}^{L-1} 5^{L-1-j} \varepsilon_j 2^{S_j}. \quad (14)$$

Hence

$$x_0 = 5^r \frac{\sum_{j=0}^{L-1} 5^{L-1-j} \varepsilon_j 2^{S_j}}{2^A - 5^L} = 5^r y_0. \quad (15)$$

Thus the scaled Diophantine cycle equation contains no new integrality condition beyond the base equation. This supplies a second direct proof of cycle preservation.

8 Orbit statistics preserved by conjugacy

Let

$$y_j = T_0^j(y), \quad x_j = T_r^j(5^r y).$$

Then $x_j = 5^r y_j$ for all j .

8.1 Capture/transient time

Let $\tau_0(y)$ denote the first iteration at which the base orbit enters one of the displayed attractors $\{1\}$ or $\{7, 9, 11\}$. Then

$$\tau_r(5^r y) = \tau_0(y). \quad (16)$$

The complete valuation sequence $(a_0, a_1, \dots)$ is identical, so the accumulated number of individual divisions by 2 over any matched finite segment is also identical.

8.2 First-descent stopping time

Independently of capture time, define the first-descent stopping time

$$\sigma_0(y) = \min\{j \geq 1 : T_0^j(y) < y\},$$

whenever such a j exists. Since multiplication by 5^r preserves strict inequalities,

$$T_r^j(5^r y) < 5^r y \iff T_0^j(y) < y.$$

Therefore

$$\sigma_r(5^r y) = \sigma_0(y). \quad (17)$$

If

$$L_0(y) = T_0^{\sigma_0(y)}(y)$$

is the first lower value, then

$$L_r(5^r y) = 5^r L_0(y). \quad (18)$$

8.3 Peaks and normalized ratios

For any matched finite trajectory segment,

$$\max_j x_j = 5^r \max_j y_j. \quad (19)$$

The raw affine values scale in the same way because $M_r(5^r y) = 5^r M_0(y)$. Thus both reduced and raw affine peaks are multiplied by 5^r . On the other hand,

$$\frac{x_j}{x_0} = \frac{y_j}{y_0}, \quad (20)$$

so every normalized peak/start ratio and every normalized trajectory profile is invariant. The iteration index at which a matched peak occurs is also unchanged.

8.4 Basin membership

For a base cycle C , let $B_0(C)$ be its basin and let $B_r(5^r C)$ be the basin of the scaled cycle within S_r . Then

$$5^r y \in B_r(5^r C) \iff y \in B_0(C). \quad (21)$$

Consequently basin counts and basin proportions are exactly preserved on corresponding finite sets. They should not be compared at the same absolute cutoff without accounting for the different scaled domains.

9 Two-adic valuation law and drift under scaling

The base map forces $a = \nu_2(M_0(y)) \geq 2$. In the natural two-adic residue model,

$$\Pr(a = k) = 2^{-(k-1)}, \quad k \geq 2,$$

and

$$\mathbb{E}[a] = 3.$$

For large y , the base logarithmic increment is approximated by

$$\log \frac{T_0(y)}{y} \approx \log 5 - a \log 2,$$

giving

$$\mathbb{E}[\Delta \log] \approx \log 5 - 3 \log 2 = \log(5/8) \approx -0.4700036292. \quad (22)$$

The geometric typical multiplier is therefore $5/8$. The ordinary expected approximate multiplier is instead

$$\mathbb{E} \left[\frac{5}{2^a} \right] = \frac{5}{6}.$$

For matched states the normalized one-step ratio is exact:

$$\frac{T_r(5^r y)}{5^r y} = \frac{T_0(y)}{y}. \quad (23)$$

Hence the scaled family inherits the same valuation distribution, the same normalized multiplicative process, the same critical average valuation $\log_2 5$, and the same drift heuristic. As in the base paper, negative expected logarithmic drift is a probabilistic interpretation and not a proof that every orbit converges.

10 Exact inverse branches and scaled inverse forests

Suppose $T_0(y) = z$ and the removed power is 2^a , $a \geq 2$. The base inverse candidates are

$$y = \frac{2^a z + 1}{5} \quad \text{or} \quad y = \frac{2^a z - 1}{5},$$

subject to integrality and the corresponding branch condition.

For the scaled system, if $T_r(x) = w$ with $x = 5^r y$ and $w = 5^r z$, then

$$x = 5^r \frac{2^a z + 1}{5} = \frac{2^a w + 5^r}{5},$$

or

$$x = 5^r \frac{2^a z - 1}{5} = \frac{2^a w - 5^r}{5}.$$

Thus every inverse edge in the base forest is multiplied by 5^r , and conversely division by 5^r sends every inverse edge in S_r back to a base edge. The entire rooted inverse forest of the fixed point and the entire rooted inverse forest of the three-cycle are therefore scaled copies, not merely forward-orbit copies.

Because the inverse-tree geometry is transported bijectively, any basin-density statement formulated intrinsically on corresponding scaled sets is equivalent to its base statement. The observed finite 72/28-type split is inherited on corresponding finite sets; no new limiting-density theorem follows from scaling alone.

11 Finite verification inherited from the base paper

The base computation exhaustively tested all positive odd starting values below 10^{10} :

5,000,000,000 starts.

Every tested orbit reached one of the two displayed attractors within the stated safety cap. The inherited base summary is:

Quantity	Base value through 10^{10}
Odd starts	5,000,000,000
Basin of 1	3,599,424,753 (71.988495060%)
Basin of $7 \rightarrow 9 \rightarrow 11 \rightarrow 7$	1,400,575,247 (28.011504940%)
Unresolved	0
Mean capture time	43.775639881800
Maximum capture time	174 at 8,793,787,045
Maximum reduced odd peak	11,134,926,214,451
Peak-holder start	9,250,534,785
Maximum raw affine peak	55,674,631,072,256
Maximum observed ν_2	34
Maximum reduced-peak/start ratio	3673.419845666274

These are not new measurements of the 5^r family. The exact theorem implies that for every fixed r , the corresponding set

$$F_r = \{5^r y : 1 \leq y < 10^{10}, y \text{ odd}\} \quad (24)$$

contains exactly 5,000,000,000 scaled starts and inherits the same basin counts, proportions, capture-time distribution, maximum capture time, and valuation sequences.

Absolute values scale. In particular, on F_r :

$$\text{maximum reduced peak} = 5^r \cdot 11,134,926,214,451,$$

attained by

$$5^r \cdot 9,250,534,785,$$

and the maximum raw affine peak is

$$5^r \cdot 55,674,631,072,256.$$

The maximum capture time remains 174, attained at the scaled start

$$5^r \cdot 8,793,787,045.$$

The maximum observed valuation remains 34, and the maximum normalized reduced-peak/start ratio remains exactly the same numerical ratio, 3673.419845666274.

The corresponding finite set lies below absolute scale $5^r 10^{10}$, but it is a structured subset S_r , not the set of all odd integers below that absolute bound.

12 Finite transfer theorem and global equivalence

Theorem 12.1 (Finite transfer theorem). *For every $r \geq 0$, every scaled start*

$$x = 5^r y, \quad 1 \leq y < 10^{10}, \quad y \text{ odd},$$

enters either the fixed point 5^r or the scaled three-cycle

$$7 \cdot 5^r \rightarrow 9 \cdot 5^r \rightarrow 11 \cdot 5^r \rightarrow 7 \cdot 5^r$$

under T_r , subject exactly to the same finite computational conditions used in the base verification.

Proof. The base computation establishes the finite statement for each corresponding y . The all-iterates identity (10) transports the entire verified trajectory by multiplication by 5^r . $\square$

Conjecture 12.2 (Scaled two-attractor convergence). *For every $r \geq 0$ and every $x \in S_r$, the orbit of x under T_r eventually enters either 5^r or the scaled three-cycle (12).*

Theorem 12.3 (Equivalence of global conjectures). *The scaled two-attractor conjecture holds for any fixed $r \geq 0$ if and only if the base two-attractor conjecture holds on S . Consequently it holds for all $r \geq 0$ if and only if it holds for $r = 0$.*

Proof. The conjugacy Φ_r transports every orbit in both directions. $\square$

Thus the parameter r creates no independent global convergence problem. It also does not solve the base problem. A divergent base orbit, if one exists, scales to a divergent orbit in every S_r ; an undiscovered base cycle scales to a cycle of the same length in every S_r .

13 What is proved and what is not

The exact algebraic results prove, for every $r \geq 0$:

- (i) $S_r = 5^r S$ is in bijection with the base domain;
- (ii) multiplication by 5^r preserves the relevant residue modulo 4;
- (iii) $M_r(5^r y) = 5^r M_0(y)$;
- (iv) the complete two-adic valuation is unchanged at corresponding steps;
- (v) T_r and T_0 are exactly conjugate;
- (vi) all iterates correspond exactly;
- (vii) branch sequences and valuation sequences correspond exactly;
- (viii) periodic orbits and minimal cycle lengths correspond bijectively;
- (ix) capture/transient times are preserved;
- (x) first-descent stopping times are preserved and first-lower values scale by 5^r ;
- (xi) absolute reduced and raw peaks scale by 5^r ;
- (xii) peak locations in iteration time and all normalized peak/start ratios are preserved;

- (xiii) basin membership and basin counts on corresponding finite sets transfer exactly;
- (xiv) inverse branches and the complete inverse forests scale exactly;
- (xv) the base finite verification through 10^{10} transfers to the corresponding scaled set through absolute scale $5^r 10^{10}$.

What is *not* proved is global convergence of every positive base orbit, nonexistence of additional base cycles, or existence and exact values of limiting basin densities. A finite scan through 10^{10} cannot exclude phenomena beyond the tested range. Since the scaled systems are conjugate to the base system, they inherit exactly this unresolved global status.

14 Reproducibility and computational verification of the identity

The supplementary archive accompanying this paper contains the L^AT_EX source, compiled PDF, a small Python verifier, a machine-readable JSON statement of the scaling rules and inherited base statistics, a README, the inherited base summary, and SHA-256 checksums.

The verifier checks, for user-selected finite ranges of r and odd y , the identities

$$\begin{aligned} M_r(5^r y) &= 5^r M_0(y), \\ \nu_2(M_r(5^r y)) &= \nu_2(M_0(y)), \\ T_r(5^r y) &= 5^r T_0(y), \end{aligned}$$

and compares several iterates. This verifier is a consistency check, not the proof. The proof is the algebraic factorization together with the facts $5^r \equiv 1 \pmod{4}$ and $\gcd(5^r, 2) = 1$.

The 10^{10} numerical evidence belongs to the base record. The present paper deliberately avoids presenting inherited trajectories as newly recomputed data.

15 Conclusion

Replacing the base constants ± 1 by $\pm 5^r$ while scaling the odd state space by 5^r produces an exact scaled copy of the reduced ternary $(5k \pm 1)/4$ dynamics. The generalized affine branches are

$$5x - 5^r, \quad 5x + 5^r,$$

and their forced first compression is of the form $(5x \pm 5^r)/4$. After complete two-adic reduction,

$$T_r(5^r y) = 5^r T_0(y)$$

for every $r \geq 0$ and every positive odd y .

The reason is exact and elementary: powers of 5 are simultaneously congruent to 1 modulo 4 and coprime to 2. The first fact preserves branch selection; the second preserves the complete two-adic valuation. Consequently every scaled trajectory is precisely the 5^r -multiple of one base trajectory.

Every scaled periodic orbit is the 5^r -multiple of a base periodic orbit with the same minimal period. The fixed point and the three-cycle therefore become 5^r and

$$7 \cdot 5^r \rightarrow 9 \cdot 5^r \rightarrow 11 \cdot 5^r \rightarrow 7 \cdot 5^r.$$

Capture times and first-descent stopping times are unchanged; first-lower values and absolute peaks scale by 5^r ; normalized peak ratios are unchanged; and the inverse forests and basin memberships are transported bijectively.

The exhaustive base verification of all five billion odd starts below 10^{10} therefore transfers, without a new billion-scale scan, to every corresponding 5^r -scaled finite domain. At the same time, the global convergence question is unchanged. Proving the two-attractor conjecture for the base system would prove it simultaneously for the whole scaled family, while any base counterexample would propagate throughout the family. The parameter r is therefore an exact scaling symmetry, not a source of independent dynamics.

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